

SUMMER INTERNSHIP PROJECT REPORT

Time-Series Analysis of Global Temperature Data

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1. Abstract

This project explores two core pillars of data science workflows: climate data engineering and unsupervised image classification. The first phase handles a time-series climate dataset tracking historical global temperature anomalies across multiple recording authorities over time. The data was filtered dynamically to isolate recent generational shifts, followed by multi-dimensional aggregations and structural pivot layouts to map monthly trends. The second phase builds an unsupervised pattern recognition architecture using the MNIST digits dataset containing 1,797 structural pixel matrices. A K-Means model initialized with ten clusters successfully grouped high-dimensional image inputs without using predefined class targets. To resolve the machine's arbitrary group indexing, a statistical mode-mapping post-processing logic was established to translate clusters into true human-understandable labels. The finalized pipeline achieved a Macro-Averaged F1 Score of 0.8628, confirming a strong precision-recall balance.

2. Introduction

In modern research, handling large-scale data requires a hybrid mastery of data cleaning, time-series aggregation, and predictive modeling. This project applies structural data engineering and automated machine learning pipelines to two real-world datasets: global historical temperatures and handwritten numerical images. The technology stack used throughout this workflow relies entirely on the Python ecosystem, specifically utilizing Pandas and NumPy for vector manipulation, SciPy for statistical distribution analysis, and Scikit-Learn for training cluster models and measuring evaluation metrics.

Prior to undertaking this analysis, comprehensive foundational training was completed during the first two weeks of the internship. The structured training modules covered:

Advanced Python programming paradigms and object properties.

Scientific data manipulation using NumPy arrays and vector math operations.

Data engineering, filtering, slicing, and feature extraction using Pandas DataFrames.

Exploratory Data Analysis (EDA) techniques and statistical data visualization.

Statistical foundation concepts including central tendencies, variances, and frequency distributions.

3. Project Objective

The systematic objectives structured for this analytical project are as follows:

- To implement dynamic, future-proof time-series filtering on global temperature records independent of ongoing dataset expansion.
- To evaluate measurement fluctuations across independent historical data recording sources (GCAG and GISTEMP).
- To aggregate temporal variations across years and individual calendar months to observe modern historical warming patterns.
- To reconstruct flat time-series records into multi-dimensional matrix layouts cross-referencing global anomalies across custom chronological grids.
- To design a robust, unsupervised K-Means clustering configuration capable of isolating structural patterns in high-dimensional image pixels without class boundaries.
- To create an automated post-processing optimization loop that resolves unsupervised string mismatch by pairing cluster distributions with actual modal digit targets.

4. Methodology

The execution architecture was carried out through an orderly multi-step data preparation and mathematical modeling framework:

Data Engineering & Cleaning Pipeline (Phase 1)

- **Ingestion:** Historical climate measurements were loaded from a flat-file environment (monthly_csv.csv) directly into an optimized Pandas DataFrame layout.
- **Dynamic Boundary Calculation:** Rather than hardcoding fixed index rules, the pipeline programmatically identifies the absolute temporal ceiling of the dataset (latest_year) and calculates a relative threshold using $(\text{latest_year} - 20) + 1$ to isolate a consistent 20-year analytical window.
- **Statistical Aggregation:** Data structures were partitioned using localized grouping constraints to measure independent means across tracking source variables, annual iterations, and calendar months.
- **Reshaping:** Flat records were pivoting-transformed into an intuitive spatial matrix map using an internal zero-fill parameter (fill_value=0) to eliminate structural sparsity and mask unrecorded periods cleanly.

Unsupervised Machine Learning Pipeline (Phase 2)

- **Feature Extraction:** Scikit-Learn's native MNIST dataset was initialized, splitting records cleanly into a structured features matrix (x) containing 64 independent structural dimensions (representing $8 * 8$ grayscale pixels) and a corresponding target list (y). No training-testing partition split was performed since clustering operates as a purely unsupervised learning technique.
- **K-Means Clustering Execution:** A clustering architecture was configured with the structural target hyperparameter locked firmly at $k=10$ to match the true distribution of numerical entities (digits 0 through 9). The model evaluated spatial distances across instances using the Euclidean distance formula

- **Modal Label Mapping Logic:** Because K-Means registers patterns by returning arbitrary integer groupings (e.g., placing actual zeroes into an unmapped category called "Cluster 3"), an automated translation algorithm was integrated. The script creates an iterative loop from 0 to 9, flags rows via boolean array masking, isolates the ground-truth target values belonging to that cluster, calculates the statistical **mode** (most frequent number), and overwrites the arbitrary cluster value with that winning identity.

5. Data Analysis and Results

Time-Series Climate Explorations

The time-series cleaning phase structured the global temperature observations effectively. Sub-setting the dataset down to the most modern 20-year window preserved high data completeness across both prime tracking branches. Grouping the data across calendar months (1 to 12) isolated specific seasonal shifts, showing an upward baseline divergence in global anomalies over modern cycles. The finalized pivot matrix structured individual tracking years sequentially as rows against month series columns, building a complete, filled chronological grid for reporting.

Clustering Classification Performance

The dimensions of the feature and target arrays used in the machine learning workflow were verified as follows:

- **Shape of Feature Matrix (x):** (1797, 64) — representing 1,797 image instances containing 64 numeric pixel variables each.
- **Shape of Target Array (y):** (1797,) — representing the matching vector of actual historical labels.

Once the automated modal alignment function renamed the machine-generated cluster outputs to match human categories, the classification alignment was validated via a **Macro-averaged F1 Score**. This metric evaluates the macro harmonic mean of precision and recall uniformly across all ten independent categories.

The finalized optimization loop returned a perfect, clean system execution output:

The F1 score is 0.8627854

An unsupervised system hitting an F1 metric of **0.8628** proves that the K-Means algorithm successfully organized the visual boundaries, pixel alignments, and contours of the handwritten characters with roughly 86.3% accuracy without ever referencing user classification targets during its core fitting cycle.

6. Conclusion

This summer research project successfully demonstrates the execution of advanced Python data pipelines from foundational data merging up to complete machine learning evaluation architectures. The historical climate phase reveals structural warming distributions over time using clear cross-referenced table manipulation. Concurrently, the unsupervised clustering component establishes that distance-based spatial algorithms can automatically discover mathematical structures within high-dimensional pixel fields. Resolving the unsupervised label-mismatch using automated modal assignments provides a practical template for real-world cluster classification, ensuring strong consistency and performance validation across data disciplines.

7. Appendices

Project Deliverables & Code Repository

- [GitHub Official Code Repository Link](#) : Includes Dataset & Assignment Code